Work Function-Based Gas-Sensors Using Doped 2D Nanomaterial (SnS_2): A DFT Study

This Major project report to the University of Calicut for the partial fulfillment of the requirements for the award of the degree of

MASTER OF TECHNOLOGY

In

NANOSCIENCE AND TECHNOLOGY

By

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PROJECT REPORT

Submitted to,

Department of Nanoscience and Technology

University of Calicut

University of Calicut

Department of Nanoscience and Technology

2022

CERTIFICATE

Certified that this is a bonafide record of the project titled

Work Function-Based Gas-Sensors Using Doped 2D Nanomaterial (SnS_2): A DFT Study

Done by

SHINTA DAVIS

of final year M.Tech in partial fulfillment of the requirement for the award of the Degree of Master of Technology in Nanoscience and Technology of the The University of Calicut during the year 2022

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DECLARATION

I, SHINTA DAVIS hereby declare that the project report entitled 'Work Function-Based Gas-Sensors Using Doped 2D Nanomaterial (SnS_2): A DFT Study' is an authentic record of the work carried out by me under the supervision and guidance of Dr. Raghu Chatanathodi, Professor, Department of Physics, National Institute of Technology Calicut, and submitted to the University of Calicut for the partial fulfillment of the requirement for the award of MTech Nanoscience and Technology, and that no part of this work has been presented before for any other degree.

SHINTA DAVIS
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NIT CALICUT
10/08/2022

DEPARTMENT OF PHYSICS

NATIONAL INSTITUTE OF TECHNOLOGY CALICUT

CERTIFICATE

This is to certify that the project report entitled **Work Function-Based Gas-Sensors Using Doped 2D Nanomaterial (SnS_2): A DFT Study**, submitted by SHINTA DAVIS of the Department of Nanoscience and Technology, the University of Calicut towards partial fulfillment of requirements for the award of the Degree of Master of Technology in Nanoscience and Technology is a bonafide record of work carried out by the student under my supervision and guidance. The contents of this project have not been submitted in exactly similar form to any other institution or University for the award of any degree or diploma.

Dr. Raghu Chatanathodi
Professor, Department of Physics
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2022

CERTIFICATE

This is to certify that the project report entitled **Work Function-Based Gas-Sensors Using Doped 2D Nanomaterial (SnS_2): A DFT Study**, submitted by **SHINTA DAVIS Reg.No:CUAUCNT007** of final year MTech, of the Department of Nanoscience and Technology, the University of Calicut towards partial fulfillment of requirements for the award of the Degree of Master of Technology in Nanoscience and Technology is a bonafide record of work carried out by the student under the supervision and guidance of Dr. Raghu Chatanathodi, Professor, Department of Physics, National Institute of Technology Calicut. The contents of this project have not been submitted in exactly similar form to any other institution or University for the award of any degree or diploma.

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ACKNOWLEDGEMENT

I wish to express my deep sense of gratitude and obligation to Dr. Raghu Chatanathodi, Professor, Computational Material Science Lab, Department of Physics, NIT Calicut, for his efficient guidance and support for this project work.

I also express my sincere gratitude to Ms. Anju Rajan, Research scholar, Department of Physics, NIT Calicut for her guidance and support throughout my project.

I take this opportunity to thank all other members of the lab who have directly or indirectly helped me to complete this project. Then, I would like to express my thanks to my friends, teachers, and family for their cooperation and support.

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Abstract

Gas sensors are extensively used in the detection of toxic and harmful gases and play a key role in conserving the quality of the environment. In this work, we model a work function-based gas sensing material. By performing first principles calculations, the adsorption of methanol and ethanol gases on doped-SnS_2 is studied to utilize its potential as a gas sensor. A stable monolayer of 2D material SnS_2 is used as the base material for the adsorption of target gas molecules. Since target gas adsorption on the monolayer is found insignificant, the monolayer is doped with metals like Pt, Ru, and metal oxide like CuO via adsorption. We investigate the binding of metal atoms at different sites on the SnS_2 monolayer and find the preferred adsorption site for all the dopants. The doped monolayer is then examined for adsorption of the target gases and the work function change ($\Delta\varphi$). The results reveal that a doped SnS_2 monolayer can be deployed as an efficient gas sensing material.

CHAPTER 1

INTRODUCTION

Gas sensors are extensively used in the detection of toxic and harmful gases and play a key role in conserving the quality of the environment. With the rise in nanotechnology, researchers are motivated to design novel gas sensors based on new 2D nanomaterials, owing to their remarkable properties. One of the best advantages that 2D material-based nano-sensors show is the high surface-to-volume ratio, which increases rates of surface reactions and also allows easy attachment of the target molecule to the sensing substrate, over conventional microsensors[16].

Methanol is a kind of flammable, volatile, and toxic vapor that is widely used as an industrial energy source. It has an obvious anesthetic effect on the human body and especially has a significant selective effect on the optic nerve and retina. To achieve safe detection of the flammable, toxic vapor of methanol, a high-performance gas sensor capable of operating at a low temperature is highly required[12].

Ethanol or ethyl alcohol is a colourless, flammable, and volatile liquid. It is a common chemical material that has wide applications in pharmacology, chemical industries, hygienic and cosmetic production, food industries and, it can be considered as one of the most important renewable resources for hydrogen (H_2) production for fuel cell applications[15].

In recent years, ethanol gas sensors have attracted more attention, due to their applications in various areas, like biomedical, chemical, and quality control of food production, environmental monitoring, indoor air quality, and determining the BrAC in the body of the drivers. BrAC is related to the level of intoxication: haziness: (0.25-0.5 mg/l, 130-260 ppm); slight drunkenness: (0.5-0.75 mg/l, 260-390 ppm); drunkenness: (0.75-1.25 mg/l, 390-650 ppm). Also, in most countries, BrAC of 130-208 ppm is a legal and permitted concentration for driving. Therefore, quick detection of ethanol gas vapours, especially at an upper than 100 ppm concentration, is considerably required[14].

[17]Transition-metal dichalcogenides (TMDs) have recently become the focus of sensing applications due to their high surface-by-volume ratio, strong chemical reactivity, and high carrier mobility, especially in two-dimensional (2D) structures. Among them, monolayer MoX_2 (X = S, Se, and Te) are first explored and reported to have strong potential for gas sensing

applications. However, an SnS_2 monolayer is another TMD that has similar structure and property to those of a MoS_2 monolayer. The SnS_2 monolayer is an indirect band gap semiconductor with a band gap of 2.60 eV[17] and superior carrier mobility of 50 $cm^2 V^{-1}s^{-1}$ which confers its application for gas sensors[9], lithium-ion batteries[13], and water splitting[5].

Moreover, through n-doping or p-doping of the SnS_2 monolayer, much more desirable properties could be obtained, like Pd doped SnS_2 monolayer for gas sensing in transformer oil[10], Ru-doped SnS_2 monolayer for VOC sensing to exploit its potential in lung cancer diagnosis[19], Pt- SnS_2 monolayer can be employed as an efficient candidate in the sensing of toxic gases like NO, NO_2 and NH_3, etc[16]. This inspired us to exploit the possibility of metal-doped SnS_2 as Methanol and Ethanol gas sensors.

[12]However, under the high surface activity provided by the high specific metal atoms, the doped pure metal atoms on the SnS_2 surface are easily oxidized in some sensor applications. By comparison, metal oxides modification demonstrates chemical stability as a dopant on SnS_2 and holds outstanding gas-sensing properties to small gas molecules. For example, CuO-doped graphene is considered to be a splendid sensor for H_2S detection; CuO-doped BNNT has ideal adsorption efficiency to C_2H_2, H_2, and CH_4; NiO-decorated ZnO nanoflowers have good performance for SO_2, SO_2F_2, and SOF_2. Due to the excellent surface activity of CuO and NiO, these metal oxides doped SnS_2 have been chosen to study the gas-sensing response to ethanol and methanol.

In this work, we model a work function-based gas sensing material. Widely studied sensing devices are transistors that depend on the work function change caused by the gas adsorption. Work function (φ) is defined as the energy which is essential to move an electron from the Fermi level to the vacuum level. The φ value differs with surfaces which means that it could represent the sensing properties because of its strong sensitivity toward the gas adsorption. In addition, the surface work function is a measurable quantity and thus is the basis of surface φ sensors[8]. The work function change (Δφ) between the target gas on the doped SnS_2 monolayer system and the doped SnS_2 monolayer is the gas sensing parameter studied in this work.

Advancements in computer technology marked the first- principle calculation as an important research tool. In recent years, first- principle calculations have been extensively used for exploring the possibility of using the excellent properties of various 2D nano-materials for gas detection. The SnS_2 material being earth-abundant and nontoxic semiconductor because of its remarkable electronic properties has aroused lots of research interest to be used as a gas sensing

substrate. The results of previous research already demonstrated that defects or dopants can tune the sensing performance of gas sensors. The various published results also show that by doping the transitional metal (TM) atoms to the SnS_2 nano-material the gas sensing performance can be effectively enhanced[16].

In this study, we explored the possibility of using metal (Pt, Ru)[15] doped SnS_2 and metal oxide(CuO, NiO)[1] doped SnS_2 as gas sensors for sensing Methanol and Ethanol gas molecules. We performed the theoretical calculations to study the impact of metal and/or metal oxide atom doping on pristine SnS_2 monolayers and also on the structural and electronic properties of doped SnS_2 monolayers, after the adsorption of gas molecules. The work function and adsorption energy have been calculated. The structural optimization and all characterization studies regarding the adsorption process were performed in a computational program VASP, employing the density functional theory (DFT) method.

CHAPTER 2

THEORY

To study the gas sensing properties of SnS_2 monolayer we should have a sufficient understanding of the adsorption at various sites and work function. These properties need to be calculated for our system. Since we are dealing with the atomic realm, we have to use quantum mechanics and should solve the many-body Schrodinger equation to get the ground state energy of the system. For such a system, the solution requires high computational effort, as the number of atoms in the system increases. In our study, we have used Density Functional Theory (DFT) calculations.

2.1 The Hamiltonian

Being very light particles, electrons cannot be described by classical mechanics. So, we use the time-independent Schrodinger equation given by,

$$H\Psi = E\Psi \tag{2.1}$$

Where H is the Hamiltonian, E is the energy eigenvalue, and the corresponding Eigen fun function-Born Oppenheimer approximation is an essential part of the solvent Schrodinger equation. Since the nuclei are massive compared to the electrons, it ignores the motion of nuclei and considers them to be stationary. Thus, an exact theory for a system of ions and interacting electrons is inherently quantum mechanical and based on solving a many-body Schrodinger equation given by,

$$\left[-\frac{\hbar^2}{2m}\sum_{i=1}^{N}\nabla_i^2 + \sum_{i=1}^{N}V(r_i) + \frac{1}{2}\sum_{i\neq j}^{N}\frac{e^2}{|r_i - r_j|}\right]\Psi = E\Psi \tag{2.2}$$

Here the first term represents the kinetic energy of each electron with mass m, the second term indicates the Coulomb potential between the nuclei and the electron and the last term is the repulsive potential energy due to electron-electron repulsion, where $\Psi = \Psi\,(r1, \ldots \ldots rN)$.

2.2 The Hartree Approximation

The ground state wave function of a many-body system can be expressed as a simple product of the orthogonal single-particle wave function in Hartree's theory. A simple Hartree product

completely neglects any correlation in the position of the electrons. All the other electrons in the system contribute to the potential.

$$\Psi_{HF}(r_1, r_2, r_3, \dots \dots .) = \Psi(r_1)^* \Psi(r_2)^* \dots\dots \quad (2.3)$$

This is called the Hartree product which converts the many-body Schrodinger equation into many one-body Schrodinger equations, which is solvable. By using the Hartree product and the variational principle, we can re-write the Schrodinger equation as,

$$\left[-\frac{\hbar^2}{2m_e}\nabla^2 + V(\vec{r}) + V_H(\vec{r})\right]\Psi_i(\vec{r}) = \varepsilon_i \Psi_i(\vec{r}) \quad (2.4)$$

Where $V_H(\boldsymbol{r}) = e^2 \int \frac{n(r')}{|r-r'|} d^3r'$ is the Hartree potential, which represents an average potential field felt by a single electron, due to all other electrons in the system.

2.3 Hartree-Fock Approximation

Now we are going to consider the indistinguishable nature of electrons. Electrons are fermions and the Hartree-Fock approximation accounts for the antisymmetry of the many-body wavefunction under the interchange of electrons. Here we choose a better wavefunction in terms of Slater determinant. The Slater determinant for N electron in terms of single electron wavefunction is given by,

$$\Psi_{HF}(r_1, r_{2,\dots} r_{N,} \sigma_1, \sigma_2, \dots, \sigma_N) = \frac{1}{\sqrt{N!}} \begin{bmatrix} \Psi_1(r_{1,}\sigma_1) & \Psi_1(r_{1,}\sigma_2) & \cdots & \Psi_1(r_N\sigma_N) \\ \vdots & & \dots & \vdots \\ \Psi_N(r_{1,}\sigma_1) & \Psi_N(r_{2,}\sigma_2) & \cdots & \Psi_N(r_N\sigma_N) \end{bmatrix} \quad (2.5)$$

Here σ denotes the spin of electrons. Applying Rayleigh-Ritz variational principle to minimize the total energy for the many-body wavefunction, we get a set of single particle Schrodinger an equation,

$$\left[\frac{\hbar^2}{2m_e}\nabla^2 + V_{ext}(r) + V_H(r)\right]\Psi_i(r) - \sum_{j=1}^{N} \delta_{\sigma i \sigma j} \int d^3r' \, \Psi_j^*(r') \frac{e^2}{|r-r'|} \Psi_j(r) = \varepsilon_i \Psi_i(r) \quad (2.6)$$

This is called the Hartree-Fock equation. When compared the Hartree's approximation, the first three terms are similar and there is an extra term. This last term is known as the exchange term, which describes the effect of exchange between electrons. The Kronecker δ over the spins σ ensures that Pauli's exclusion principle is obeyed. Thus, we see that in Hartree-Fock approximation the energy of like spins is thus reduced by the presence of an exchange hole

around every electron, while for unlike spins, the lowering of energy through an exchange is not available.

2.4 Density Functional Theory

The entire field of density functional theory (DFT) rests on two fundamental mathematical theorems proved by Kohn and Hohenberg and the derivation of a set of equations by Kohn and Sham in the mid-1960s.

Theorem 1: *The ground state energy of a quantum mechanical system is a unique function of electron density.*

Using the DFT method, instead of dealing with many-body wavefunction Ψ(ri), one could deal with problems that involve the total density of electrons n(ri). This theorem tells us about the existence of ground-state energy functional but does not provide any information about the form of the functional.

Theorem 2: *The electron density that minimizes the energy of the overall function is the true electron density corresponding to the full solution of the Schrodinger equation.*

The energy functional can be written as,

$$\mathrm{E}[\{\Psi \mathrm{i}\} = \mathrm{E}_{known}[\{\Psi \mathrm{i}\}] + \mathrm{E}_{XC}[\{\Psi \mathrm{i}\}] \quad (2.7)$$

$$\mathrm{E}[n(r)] = \mathrm{E}_{known}[n(r)] + \mathrm{E}_{XC}[n(r)] \quad (2.8)$$

Where E_{known} is given by,

$$\mathrm{E}_{known}[n(r)] = T_e[n(r)] + V_{e-n}[n(r)] + V_{n-n}[n(r)] \quad (2.9)$$

Here, the terms are in order, the electron kinetic energies, the Coulomb interactions between the electrons and the nuclei, the coulomb interactions between the pair of electrons, and the Coulomb interaction between the pair of nuclei. The other term in the complete energy functional, $\mathrm{E}_{XC}[n(r)]$ is the exchange-correlation functional which includes all the quantum mechanical effects that are not included in the known terms.

2.5 Kohn-Sham Equations

It is a very difficult task for solving the many-body Schrodinger equations. This difficulty was solved by Kohn and Sham, who showed that the task of finding the right electron density can be expressed in a way that involves solving a set of equations in which each equation only

involves a single electron. These equations are based on the assumption that the exact ground state density can be represented by the ground state density of an auxiliary system of non-interacting particles. The Kohn-Sham equations have the form,

$$\left[-\frac{\hbar^2}{2m_e}\nabla^2 + V(\vec{r}) + V_H(\vec{r}) + V_{XC}(\vec{r})\right]\Psi_i(\vec{r}) = \varepsilon_i\Psi_i(\vec{r}) \qquad (2.10)$$

The solution of the Kohn–Sham equations is single-electron wave functions that depend on only three spatial variables, $\Psi i(\vec{r})$. On the LHS of the Kohn–Sham equations, there are three potentials, V, V_H, and V_{XC}. V defines the interaction between electrons and the collection of atomic nuclei. V_H is the Hartree potential given by,

$$V_H(\boldsymbol{r}) = e^2 \int \frac{n(r')}{|r-r'|} d^3r' \qquad (2.11)$$

V_{XC} $(\vec{r})$ is the exchange-correlation potential that can be defined as a functional derivative of exchange-correlation energy,

$$V_{XC}(\vec{r}) = \frac{\delta E_{XC}(r)}{\delta n(r)} \qquad (2.12)$$

However, to solve the Kohn–Sham equations, we need to define the Hartree potential, and to define the Hartree potential we need to know the electron density. But to find the electron density, we must know the single-electron wave functions, and to know these wave functions we must solve the Kohn–Sham equations. Thus, the problem can be solved only in an iterative way. First, we have to define an initial, trial electron density, n(r) and solve the Kohn –Sham equations defined using this trial electron density to find the single-particle wave functions, $\Psi i(\vec{r})$. Then calculate the electron density defined by the Kohn –Sham single-particle wave functions and compare it with the calculated energy density. Repeat these procedures unless the two densities converge within desired criteria.

2.6 Exchange-Correlation Functional

The use of the Kohn-Sham equation is restricted since it involves the exchange-correlation energy, which is a functional of the density $E_{XC}[n(r)]$. The exact form of exchange correlation is unknown and many approximations have been used for E_{XC}. The accuracy of the DFT calculation depends on the quality of the exchange-correlation energy approximation used. Some of the exchange-correlation energy approximations are local density approximation

(LDA), generalized gradient approximation (GGA), higher order gradient or meta GGA, and hybrid orbitals.

2.6.1 Local Density Approximation (LDA)

It was noted that in the case of the uniform electron gas, exchange correlational functional can be derived exactly. In this situation, the electron density is constant at all points in space; that is, n(r) = constant. Though this situation appears to be of limited value in any real material, the uniform electron gas provides a practical way to use the Kohn–Sham equations. Kohn-Sham pointed out that molecules can often be considered as close to the limit of the homogeneous electron gas. In that limit, it is known that the effects of exchange and correlation are local, and they proposed making the local density approximation (LDA), in which the exchange-correlation energy is simply an integral over all spaces with the exchange-correlation energy density at each point assumed to be same as in a homogeneous electron gas with that density.

$$E_{XC}[n] = \int n(r) \in_{XC} [n] dr \qquad (2.13)$$

The LDA method is an exact DFT method for the special case of the uniform electron gas, except for small differences depending on the interpolation formula chosen for the correlation energy. For molecular systems, the LDA approximation underestimates the exchange energy by ~10%, thereby creating errors that are larger than the whole correlation energy. Electron correlation is overestimated, often by a factor close to 2, and bond strengths are as a consequence overestimated, often by ~100 KJ/mol

2.6.2 Generalized Gradient Approximation (GGA)

The best-known class of functional after the LDA uses information about the local electron density and the local gradient in the electron density. This approach defines a gradient corrected functional or Generalized Gradient Approximation (GGA). There are a large number of distinct GGA functionals. The functionals used for calculations in this work which involve, solids are the TDDFT2.

2.6.3 Hybrid Or Hyper GGA Methods

Hybrid functionals are a type of approximations to the exchange-correlation energy functional in which a portion of exact exchange from Hartree-Fock theory is incorporated into exchange and correlation from other sources. However, these functionals do not contain the correlation term. It is noted that in the local density description the self-interaction term cancels mutual

between exchange and correlation terms. The combination of these two theories is known as hybrid functional.

2.7 Basis Sets

For solving integral differential equations iteratively using different computational schemes, the linear combination of atomic orbital of KS molecular orbital has been adopted; i.e., the molecular orbital is expanded as the linear combination of a computationally accessible set of predefined basis functions. A basis set is a set of functions that are combined in linear combinations to create a molecular orbit. For convenience these functions are typically atomic orbitals centered on atoms, but can theoretically be any function; plane waves are frequently used in materials calculations. The two types of basis functions commonly used in electronic structure calculations: are Slater Type Orbitals (STO) and Gaussian Type Orbitals (GTO). Plane-wave basis functions are ideal for describing delocalized slowly varying electron densities, such as valence bands in metal. The core electrons, however, are strongly localized around the nuclei, and the valence orbitals have several rapid oscillations in the core region to maintain orthogonality. Describing the core region adequately thus requires a large number of rapidly oscillating functions.

2.8 Pseudopotential

The idea behind pseudopotential is to replace the strong coulombic potential of the nucleus and the tightly bound core electron with an effective ionic potential acting on valance electrons. Pseudopotential can be generated in an atomic calculation and used to compute properties of valence electrons in molecules or solids since the core states remain almost unchanged. The most commonly used pseudopotentials are Norm conserving pseudopotentials and Ultra-Soft pseudopotentials.

2.9 Projector Augmented Wave

The Projector Augmented Wave (PAW) method combines both the augmented wave method and pseudopotential method. The augmented wave method divides all-electron potential into an atom-like partial wave expansion inside the core region and into plane wave expansion for the region outside the core radius between the atoms. It considers all electrons, unlike the pseudopotential method, it considers all electrons in the system.

2.10 Adsorption Of A Target Gas By SnS_2 Monolayer

The sensing of methanol and ethanol gases by Pt doped SnS_2, Ru doped SnS_2, and CuO doped SnS_2 is studied by calculating the adsorption energy, work function, and density of states.

2.10.1 The adsorption energy (E_{ads})

The adsorption energy (E_{ads}) is used to find the energy of interaction between gas molecules and the sensing system and is calculated by the given equation:

$$E_{ads} = E_{(gas+sub)} - E_{gas} - E_{(sub)} \quad (2.14)$$

where the energy of the whole system was expressed by $E_{(sub + gas)}$ and the energy of the substrate structure and single target molecule was represented by E_{sub} and E_{gas}, respectively. A high negative value of adsorption energy indicates exothermic and energetically stable.

2.10.2 The work function(φ)

Work function () is the energy essential to move an electron from the Fermi level to the vacuum level. The φ value differs with surfaces which means that it could represent the sensing properties because of its strong sensitivity toward the gas adsorption.

work function can be calculated by using the formula below:

$$\varphi = E_{fermi} - E_{vacuum} \quad (2.15)$$

where E_{fermi} is the energy at the fermi level of the substrate. E_{vacuum} is the vacuum energy level.

CHAPTER 3

COMPUTATIONAL METHOD

3.1 COMPUTATIONAL CODE

Here we have used the Vienna ab-initio simulation package (VASP) code[2,3] for electronic structure calculations. VASP is a DFT-based electronic structure calculation code for solids and molecules using a plane wave basis. In the case of finite, nonperiodic system-like molecules we can use VASP by constructing a supercell of that molecule by placing that molecule in a cubic box representing the unit cell. As we have seen before, Besides the local density approximation (LDA) exchange correlation, there are mainly two kinds of generalized gradient approximation (GGA) functionals implemented in VASP - PW91 and Perdew Burke and Emzerhof (PBE). We have used the PBE as a reference functional for all our calculations of adsorption of target gases and further used one of the available vdW functionals in VASP, the vdW-D2 functional, to understand the Van der Waals force and long-range interactions, as proposed by Grimme, to observe the effects of vdW interactions on the work function of doped SnS_2 monolayer. For visualizing structural form, we have made use of VESTA12[7] as well as Virtual Nano Lab (VNL).

3.2 COMPUTATION DETAILS

A lattice constant of 3.687 is determined for the SnS_2 monolayer unit cell by optimizing the total energy concerning different lattice parameters. The monolayer unit cell lattice parameter computed by using various functionals is in good agreement with experimental data. For the unit cell, the plane wave cut-off energy is fixed to 360 eV and the Brillouin zone is sampled using an 8x8x8 Monkhorst- Pack grid[4]. Here we made an SnS_2 monolayer with 2 layers of S at the top and bottom and Sn at the middle layer, and these surfaces are modeled as 3×3×1 supercells with a vacuum layer of 25 A^0 along the Z direction for avoiding interaction with periodic monolayers. The monolayer contains 18 S atoms and 9 Sn atoms which is a total of 27 atoms.

The adsorption energy of target gases on a doped SnS_2 monolayer is calculated using the formula:

$E_{ads} = E_{(gas+sub)} - E_{gas} - E_{(sub)}$

where the energy of the whole system was expressed by $E_{(gas+sub)}$ and the energy of the substrate structure and single target molecule was represented by E_{sub} and E_{gas}, respectively. A high negative value of adsorption energy indicates exothermic and energetically stable.

The work function can be calculated by using the formula below:

$Ø = E_{fermi} - E_{vacuum}$

where E_{fermi} is the energy at the fermi level of the substrate. E_{vacuum} is the vacuum energy level. E_{fermi} and E_{vacuum} are calculated via running VASP code. From the OUTCAR file, the E_{fermi} value is obtained. The code will give the E_{vacuum} value or by plotting a graph between average potential and distance which are obtained by running the code we can find the E_{vacuum}

CHAPTER 4

RESULTS AND DISCUSSION

In this study we investigated the sensing properties of metal (Pt, Ru) and metal oxide (CuO)-doped SnS_2 monolayers towards methanol and ethanol gases, that is adsorption of ethanol and methanol gas molecule on doped SnS_2 monolayer using GGA functional DFT-D2.

4.1 The Optimized SnS_2 monolayer

Initially, we converged the plane wave DFT calculation concerning energy cut-off that we fixed as 360eV then the size of k point mesh for Brillouin zone integration, which is fixed at 8×8×8. These two converged parameters are then used for optimizing the lattice parameter. We have determined the lattice parameter and it is comparable with experimentally observed values.

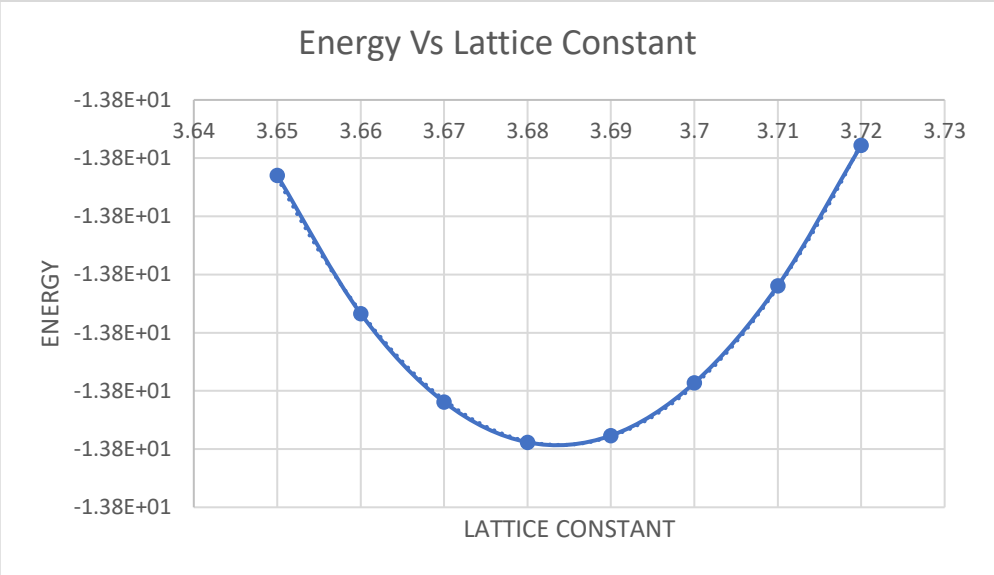
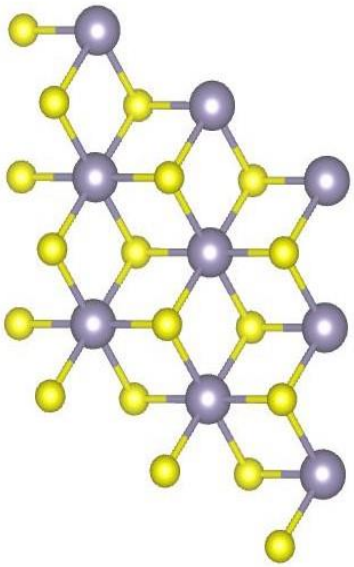
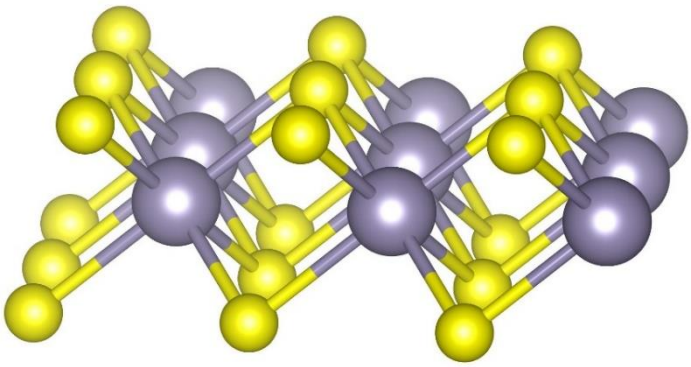


Fig.4.1: Plot between total energy and lattice constant.

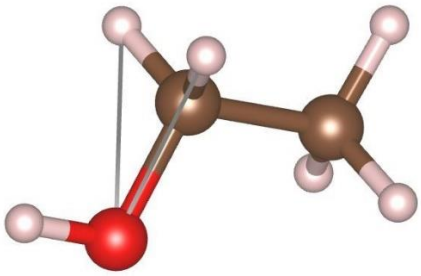
The geometric structure of all individual gas molecules and SnS_2 monolayer is initially relaxed to achieve minimum energy criteria.Fig.4.2(a-c). The optimized lattice constant and the bond length between Sulphur and Tin atom (S-Sn) of bare SnS_2 monolayer are found to be 3.68 Å and 2.59 Å, respectively[16].



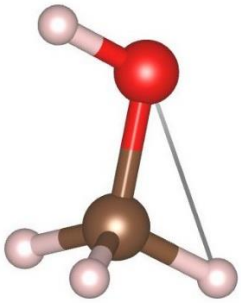
(a).Optimized SnS_2 monolayer top view



(b).Optimized SnS_2 monolayer side view



(c). Optimized ethanol molecule



(d).Optimized methanol molecule

Fig.4.2: Optimized structures of SnS_2 monolayer and target gas molecules.

4.2 Adsorption of target gas on SnS_2 monolayer

To explore the interaction between neutral surface and gas molecule we employed the 3×3×1 supercell of 2D SnS_2 monolayer with a vacuum of 25 Å to provide an adequate large interface to interact with studied gas molecules. To attain the most stable configurations of studied gas molecules adsorbed on the nano-layers, a relaxed gas molecule was initially placed at the surface and then the constructed structure was fully optimized[8].

4.2.1 The ethanol and methanol gas molecules

A relaxed ethanol fig.4.2.(c) the molecule is placed on the different sites of the relaxed SnS_2 monolayer and the structure was fully optimized. The adsorption energy (E_{ads}) of the optimized structure is 2.625eV and the adsorption energy of methanol fig.4.2.(d) on SnS_2 monolayer after

optimization is 2.703eV, which are positive values, mean poor chemical stability or no adsorption.

4.3 Doping of metals on the SnS_2 Monolayer

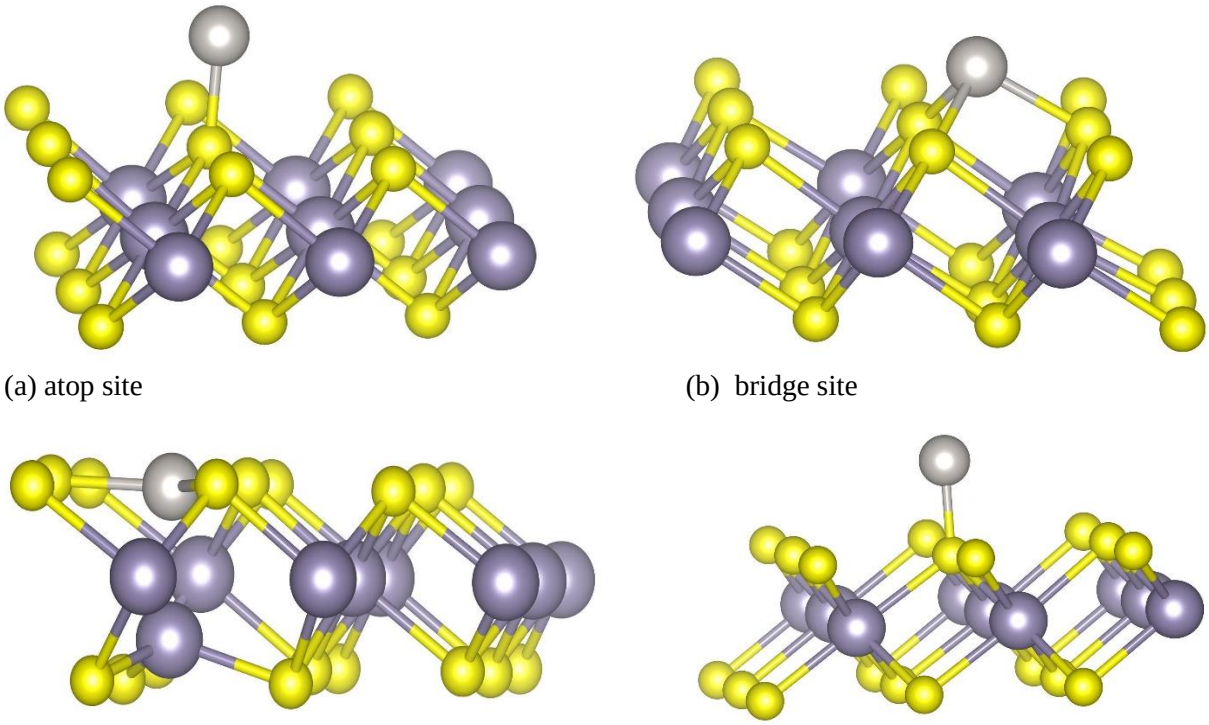
Since the E_{ads} of the optimized structure of the SnS_2- gas molecule is very low, a dopant is introduced on the SnS_2 monolayer to improve the adsorption and thereby chemical stability of the structure. In this work, we doped the SnS_2 monolayer with metals Platinum (Pt)[16] and Ruthenium (Ru)[17] which are reported in sensing gases like NO, NO_2, and CH_3OH, C_6H_6 respectively.

4.3.1 Potential Sites

There are 4 possible doping sites on SnS_2 surfaces. They are termed as the atop fig.4.2(a) (adsorption of the molecule directly above an S atom on the top layer), bridge fig.4.2(b) (adsorption of the molecule between two S atoms), hcp fig.4.2(c) (adsorption of the molecule directly above an Sn atom on the second layer), and hollow fig.4.2(d) (adsorption of the molecule directly above an S atom on the third layer). However, in the four sites, the hcp site shows the maximum adsorption with a high E_{ads} value, but as shown in the figure due to adsorption the structure is fragmented so we could not count this as a stable structure of SnS_2 monolayer. So we have considered only the other three possible sites for further discussion.

Potential sites	Pt-SnS_2 E_{ads} (eV)	Ru-SnS_2 E_{ads} (eV)
Atop	-3.51	-2.11
Bridge	-4.48	-4.46
hcp	-6.02	-5.79
Hollow	-1.69	-4.46

Table.4.1. E_{ads} of the potential site on SnS_2 monolayer



(a) atop site (b) bridge site

(c) hcp site (d) hollow site

Fig.4.2. Pt doped on potential sites on SnS_2 monolayer

4.3.2 Pt Doping Behavior on the SnS_2 Monolayer

We defined the adsorption of one Pt atom on the pristine SnS_2 monolayer to form a Pt−SnS_2 counterpart. The parameter of adsorption energy (E_{ads}) was used to evaluate the chemical stability of the Pt−SnS_2 monolayer, the E_{ads} is calculated using the equation given below and tabulated. Table 4.1

$$E_{ads} = E_{Pt\text{-}SnS_2} - [\, E_{SnS_2} + E_{Pt} \,]$$

As the above mentioned from the four doping sites we considered the three possible sites bridge, atop and hollow. After full optimization, the most stable configurations for Pt doping are through the bridge sites, which possess an E_{ads} of -4.48 eV which is very high compared to the other two sites. However, after the optimization of Pt doping on the hollow site, the Pt dopant experiences some displacement, making the structure the same as that of the atop site but with a very low E_{ads} value.

The calculated negative value of adsorption energy shows the energy stability of Pt -doped SnS_2 sensing system[17]. After doping, the Pt adatom is captured by the surrounding S atoms,

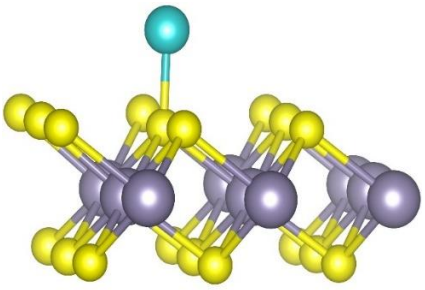
forming Pt−S bonds accordingly with bond lengths of 2.328 Å, 2.331 Å, and 2.328 Å, which are slightly shorter than the sum of covalent radii of Pt and S atoms[7,11]. This finding points to the strong binding force in the Pt−S bond with high chemical stability and the change in bond parameters shows the degree of structural deformation due to doping.

4.3.3 Ru Doping Behavior on the SnS_2 Monolayer

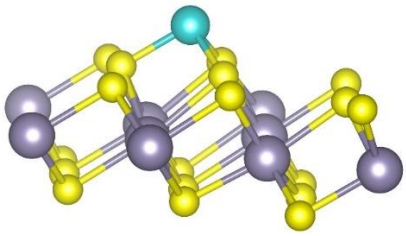
We defined the adsorption of one Ru dopant[17] on the pristine SnS_2 monolayer to form a Ru−SnS_2 counterpart. The parameter of binding energy (E_{ads}) was used to evaluate the chemical stability of the Ru−SnS_2 monolayer, the E_{ads} is calculated using the equation given below and tabulated. Table 4.1

$E_{ads} = E_{Ru\text{-}SnS2} - [\, E_{SnS2} + E_{Ru}]$.

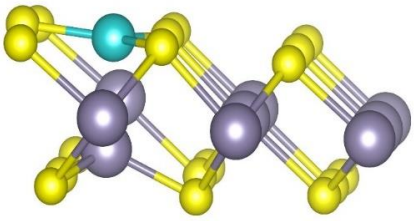
In the case of Ru doping four doping sites were considered, namely, atop, bridge, hcp, and hollow. However, just like in the case of Pt doping the hcp site shows very large adsorption and structural distortion which led us to the omission of the structure. So, after full optimization, the most stable configuration for Ru doping is through the hollow, which possesses high negative E_{ads} of -4.46 eV compared with atop site. However, after the optimization of Ru doping on the bridge site, the Ru dopant experiences some displacement, making the structure somewhat similar to the hollow site but with distortion and a slightly low E_{ads} value.



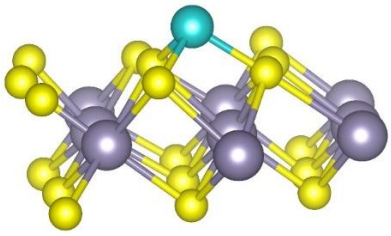
(a) atop site



(b) bridge site



(c) hcp site



(d) hollow site

Fig.4.3. Ru doped on potential sites on SnS_2 monolayer

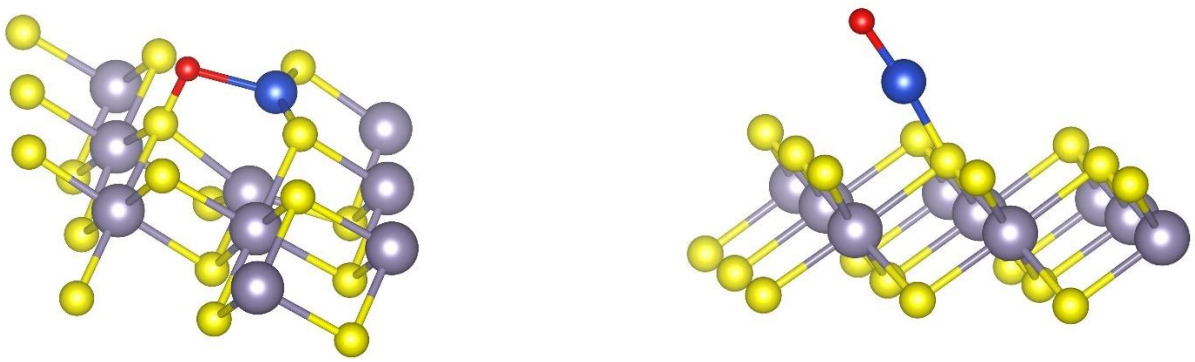
The high negative Eads value means that Ru doping on the SnS_2 monolayer through the hollow site is quite thermodynamically favorable.[17]After doping, the Ru adatom is captured by the surrounding S atom, forming Ru−S bonds accordingly with bond lengths of 2.171 Å,2.178 Å, and 2.171 Å, which are shorter than the sum of covalent radii[7,10] of Ru and S atoms. This finding manifests the strong binding force in the Ru−S bond with high chemical stability.

4.4 Metal Oxide doping on SnS_2 Monolayer

Under the high surface activity provided by the high specific metal atoms, the doped pure metal atoms on the SnS_2 surface are easy to be oxidized in the sensing applications. By comparison, metal oxides modification demonstrates chemical stability as a dopant on SnS_2 and holds outstanding gas-sensing properties to small gas molecules. Due to the excellent surface activity of CuO, the metal oxides doped SnS_2 have been chosen to study the gas-sensing response to ethanol and methanol[1].

4.4.1 CuO doping behavior on SnS_2 Monolayer

From the doping of CuO two optimized structures are obtained. fig.4.4. [1]In the first structure fig.4.4(a) the Cu–O bond is elongated to 2.176 Å during the doping process. Cu and O atoms are trapped in active sites that can be doped, simultaneously forming three bonds with the SnS_2 surface. S-O bond and two Cu-S have the bond length of 1.535 Å and 2.155 Å respectively, with an Eads of –3.061eV. In the second structure fig.4.4(b) the CuO molecule is not long parallel to the SnS_2 monolayer, but, a Cu-S bond is formed which measures 2.121 Å with an E_{ads} of -1.727eV. From the $E_{ads,}$ it is clear that the first structure is the more stable one because of its high negative value.



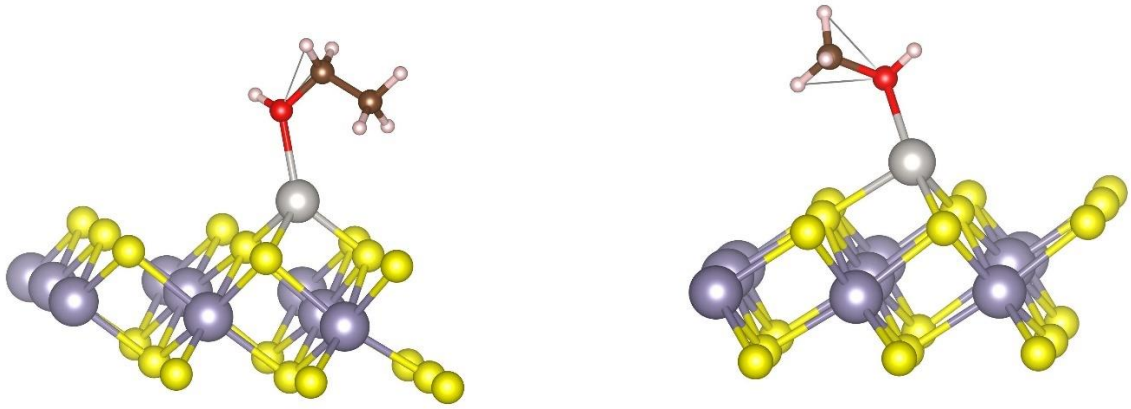
(a) Horizontal orientation of CuO (b) Vertical orientation of CuO

Fig.4.4. CuO doped on SnS_2 monolayer

4.5 Adsorption of target gases on doped SnS_2 monolayer

4.5.1 Interaction of Pt-SnS_2 sensing system with target gases

The most stable configuration of target gas molecules adsorbed on Pt doped SnS_2 system is given in the figure(4.5). The corresponding adsorption properties - adsorption energy and bond length are tabulated in Table 4.2.



(a) Ethanol Adsorbed on Pt-SnS_2 monolayer (b) Methanol Adsorbed on Pt-SnS_2 monolayer

Fig.4.5. Stable structures of target gas molecules on Pt-doped SnS_2 system.

Target gas	Pt-O Bond Length(Å)	Adsorption Energy (E_{ads}) (eV)
Ethanol	2.33	-0.75
Methanol	2.25	-0.74

Table. 4.2: Adsorption properties of target gas-Pt doped SnS_2 System

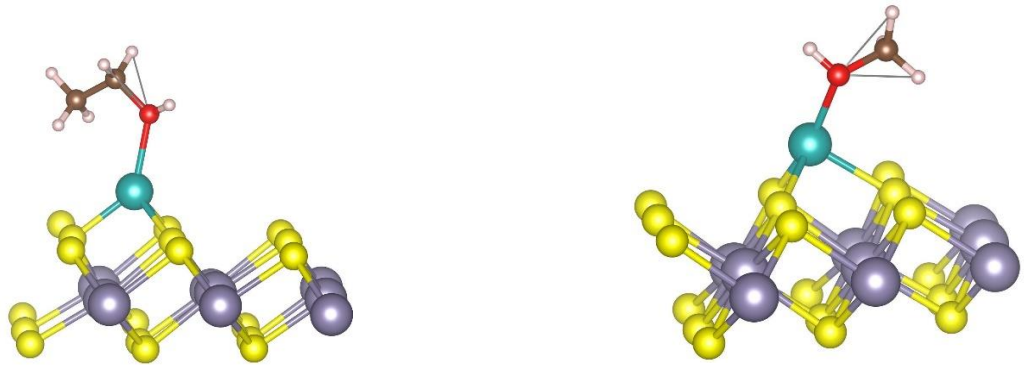
In the case of the ethanol molecule, fig.4.5(a) structural deformation is evident after optimization since the ethanol molecule is no longer parallel to the Pt-SnS_2 sensing system, which shows a geometric activation of the adsorbed gas molecules. From the two optimized structures obtained, the structure with Pt-O bond in which the O atom of ethanol is captured by the Pt dopant of the Pt-SnS2 sensor with a bond length of 2.33 Å, has the highest E_{ads} value of -0.75eV, than Pt-C bonded structure. It means that the structure with a Pt-O bond is the most stable configuration due to its high negative E_{ads} value.

The methanol molecule also gave two optimized structures with structural deformation, which shows a geometric activation of the adsorbed gas molecule. It acts in the same fashion as

ethanol gas. The most stable configuration fig.4.5(b) is the one in which Pt captured an O atom of methanol and formed a Pt-O bond with a length of 2.25 Å and an Eads of – 0.74eV.

4.5.2 Interaction of Ru-SnS2 sensing system with target gases

The most stable configuration of target gas molecules adsorbed on the Ru-SnS_2 system is given in figure.4.6. The corresponding adsorption properties - adsorption energy and bond length are tabulated in Table 4.3.



(c)Ethanol adsorbed on Ru-SnS_2 monolayer (b) Methanol adsorbed on Ru-SnS_2 monolayer

Fig.4.6. Stable structures of target gas molecules on Ru-doped SnS_2 system

Target gas	Ru-O Bond Length(Å)	Adsorption Energy (E_{ads}) (eV)
Ethanol	2.23	-1.29
Methanol	2.23	-1.16

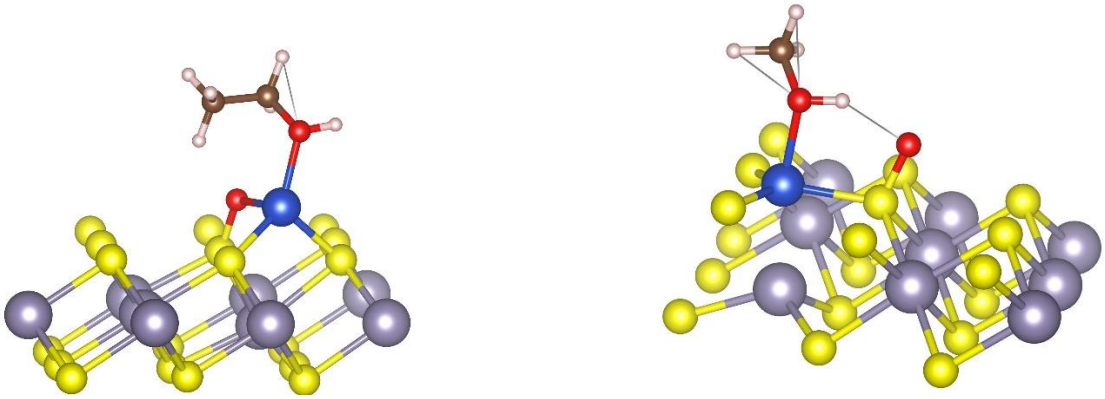
Table. 4.3(a): Adsorption properties of target gas-Ru doped SnS_2 System

We can find that the ethanol molecule prefers to be parallel to the Ru−SnS_2 surface on the top of the Ru dopant. However, the deformation of ethanol is evident after adsorption, which is no longer a plane molecule on the Ru− SnS_2 monolayer, implying the geometric activation of the adsorbed gas molecules. Besides, the O atom is captured by the Ru dopant, forming a new Ru−O bond measured as 2.23 Å. That is, the Ru dopant exerts a strong binding force upon the O atom of ethanol, which therefore leads to the large E_{ads} of -1.29 eV.

Methanol also acts in the same fashion as ethanol which shows a deformation in structure which implies a geometric activation of methanol. O atom of methanol was captured by Ru dopant of Ru-SnS_2 sensing system by forming a Ru-O bond which measured as 2.23 Å with an E_{ads} of -1.16 eV.

4.5.3 Interaction of CuO-SnS_2 sensing system with target gases

The most stable configuration of target gas molecules adsorbed on the CuO-SnS_2 monolayer is displayed in Fig. 4.6. The O atom of ethanol is captured by the Cu ion of CuO dopant and formed a Cu-O bond measured as 2.044 Å with an Eads of -0.706 eV, a pretty good value which shows the strong bonding of ethanol on CuO-SnS_2 monolayer.



(a) Ethanol Adsorbed on CuO-SnS_2 monolayer (b) Methanol Adsorbed on CuO-SnS_2 monolayer

Fig.4.6. Stable structures of target gas molecules on CuO-doped SnS_2 system.

In the case of methanol adsorption, the O atom is captured by Cu of CuO dopant to form a Cu-O bond which is 2.139 Å between the O atom of methanol and Cu of dopant, but breaks the Cu-O bond of metal oxide dopant and resulting high E_{ads} value of -0.936eV. This shows the thermodynamical stability of the CuO-SnS_2 System after target gas adsorption.

4.6 Work function analysis

Surface work function changed transistors which depend on the work function change caused by the gas adsorption is a promising method in gas sensing. Work function (φ)fig.4.7 is the energy essential to move an electron from the Fermi level to the vacuum level. The φ value differs with surfaces which means that it could represent the sensing properties because of its strong sensitivity toward gas adsorption. In addition, surface φ is a measurable quantity and thus is the basis of surface work function sensors[8].

Work function (φ) is given by,

$$\varphi = E_{vacuum} - E_{fermi}$$

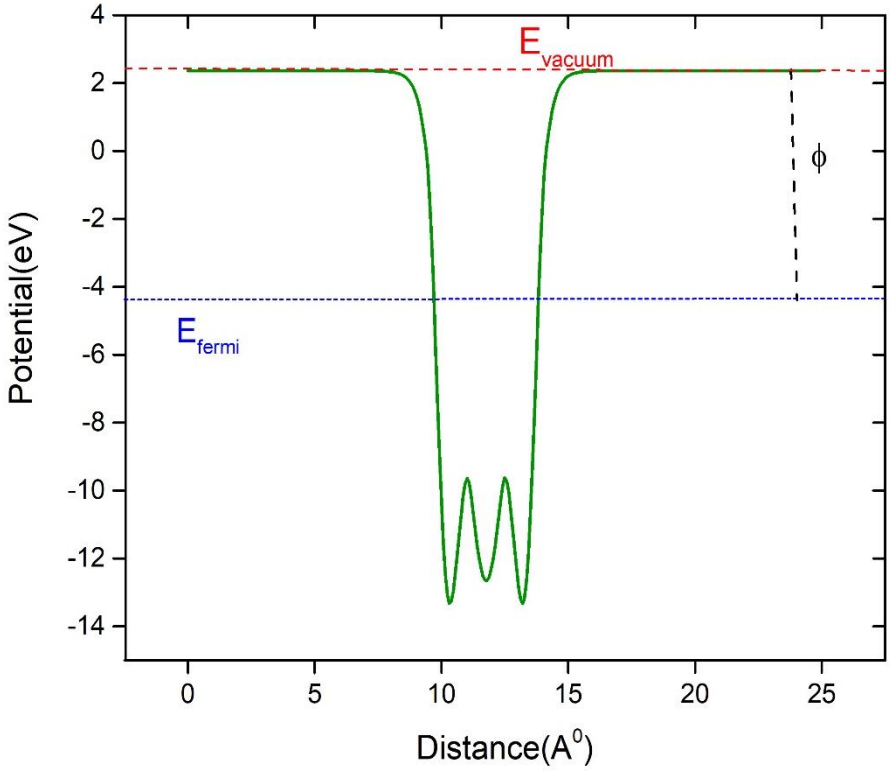
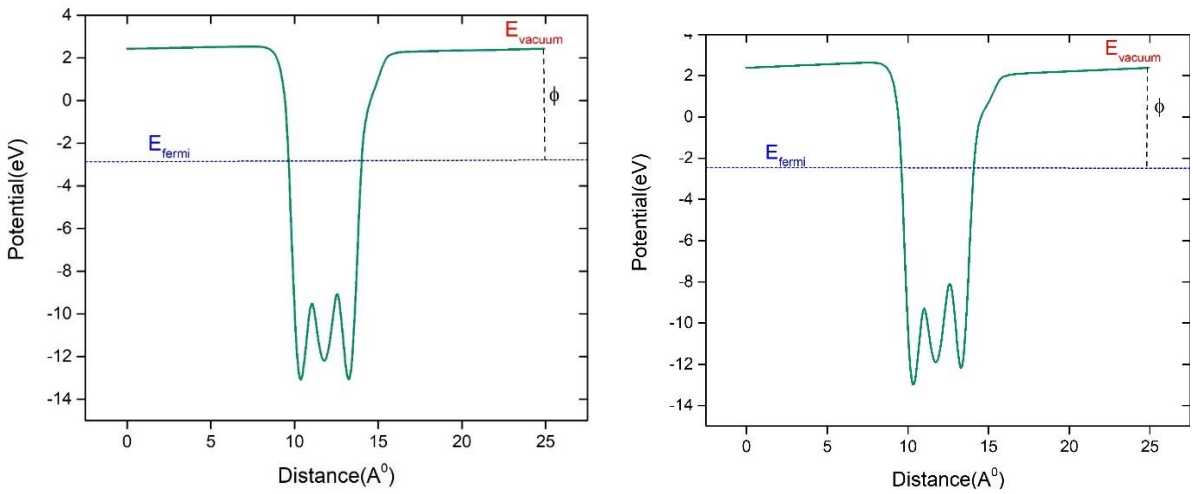


Fig.4.7. Plot of the work function of SnS_2 monolayer.

The Pt-doped SnS_2, Ru-doped SnS_2, and CuO-doped SnS_2 sensing systems are work-function change sensors. The change in the work function of the doped SnS_2 monolayer when the target gas is adsorbed is calculated and analyzed. The work function changes of Pt-doped SnS_2, Ru-doped SnS_2, and CuO-doped SnS_2 before and after ethanol and/or methanol adsorption are calculated from the graphs figure 4.8 and fig.4.9 and tabulated. Table. 4.4.



(a) Pt on SnS_2 monolayer (b) Ru on SnS_2 monolayer

Fig.4.8. Plot of the Work function of metal doped SnS_2.

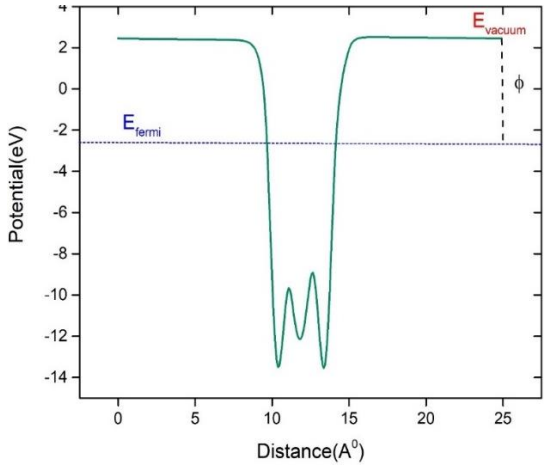
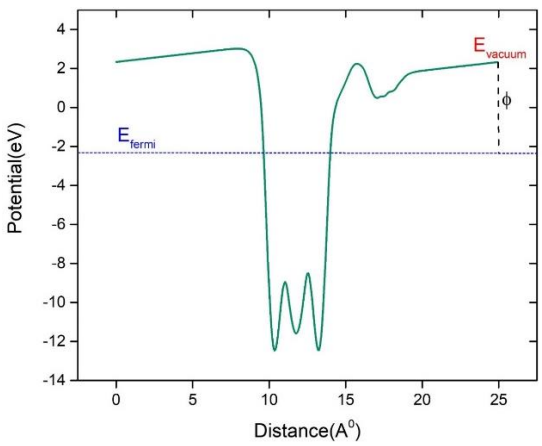
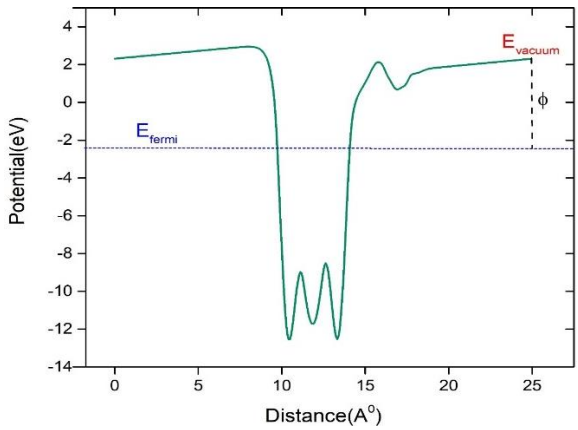


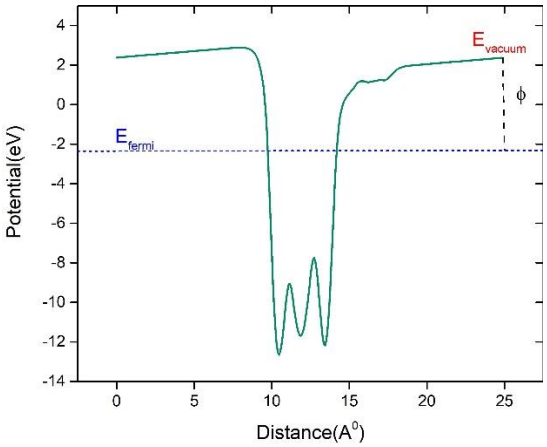
Fig.4.8. (c)Plot of the work function of CuO doped SnS_2.



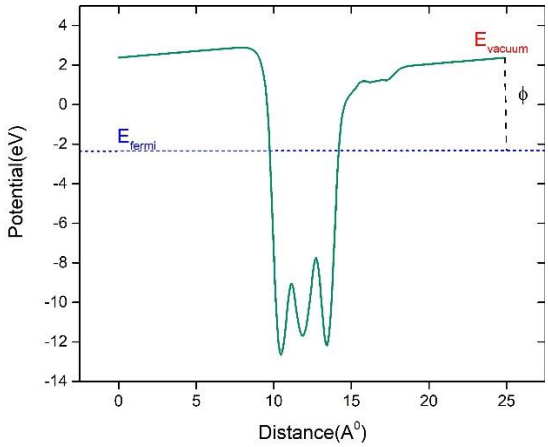
(a) Ethanol on Pt-SnS_2 system



(b) Methanol on Pt-SnS_2 system

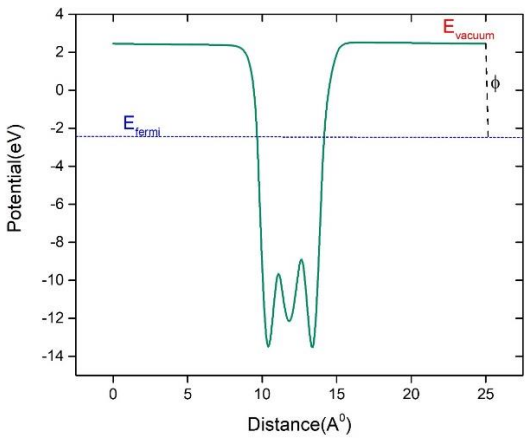


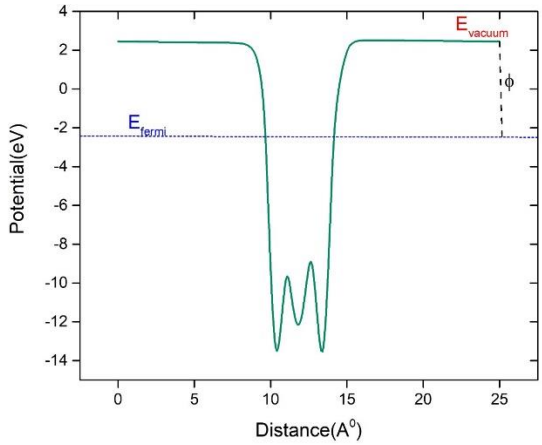
(c) Ethanol on Ru-SnS_2 system



(d) Methanol on Ru-SnS_2 system

Fig.4.9 Plot of the work function after target gas adsorption on metal doped SnS_2 monolayer.





(a)Ethanol on CuO-SnS_2 system (b) Methanol on CuO-SnS_2 system

Fig.4.10 Plot of the work function after target gas adsorption on metal oxide doped SnS_2 monolayer

	Pt-SnS_2			Ru-SnS_2			CuO-SnS_2		
Target gas	φ (eV) Before	φ (eV) After	Δφ(eV)	φ (eV) Before	φ (eV) After	Δφ(eV)	φ (eV) Before	φ (eV) After	Δφ (eV)
Ethanol	5.4834	4.7035	0.7799	4.8517	4.7246	0.1271	5.2838	4.9643	0.3195
Methanol	5.4834	4.6576	0.8258	4.9822	4.7848	0.1974	5.2838	4.9643	0.3195

Tabel.4.4 Work function before and after adsorption of target molecules.

After adsorption of ethanol gas, the work function of Pt doped SnS_2 sensing system shows a decrease of 0.7799 eV which is greater than 0.40 eV[8] indicating the Pt doped SnS_2 monolayer is very sensitive toward the ethanol gas. And in the case of methanol adsorption, the work function decreases by 0.8258 eV which is also greater than 0.40eV, indicating good sensitivity of substrate toward methanol gas.

Ru doped SnS_2 shows a decrement of 0.1271eV in work function after the ethanol adsorption which is also a good sensitivity indicator. Ru doped SnS_2 shows a work function decrement of 0.1974 eV after adsorption of methanol which also indicates a sensitivity toward methanol.

CuO doped SnS_2 shows a decrease of 0.3195eV in work function after ethanol capture and in the case of methanol also, it is 0.3195eV, which shows the sensing capacity of CuO-SnS_2 sensors.

The tabulated data shows that Pt-doped SnS_2, Ru-doped SnS2, and CuO-doped SnS_2 can be successfully used as good ethanol and methanol sensors.

CHAPTER 5

CONCLUSION

We successfully modelled work function-based gas sensing systems by doping SnS_2 monolayer with metals and metal oxide. The work function change we obtained especially from the metal-doped SnS_2 system after adsorption of target gases is very appreciable values. Work function change from metal oxide doping is also notable as compared with previously reported values[8]. Both metal and metal oxide doped SnS_2 systems are very sensitive toward the target gases ethanol and methanol, they show significant change/decrease in work function after adsorption of target gases, which proves their potential as promising candidates in gas sensing applications.

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